DAILY PRACTICE CALENDAR

Algebra 2 Competition Math (Grade 9) • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This course follows Geometry Competition Math: it builds the algebra, trigonometry and counting for **AMC 10** and gives a first look at **AMC 12**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Problem-Solving Toolkit for AMC 10
2	Aug 23–Aug 27	Functions: Domain, Range & Composition
3	Aug 30–Sep 3	Inverses & Graph Transformations
4	Sep 6–Sep 10	Absolute Value & Piecewise Functions
5	Sep 13–Sep 17	Remainder & Factor Theorems
6	Sep 20–Sep 24	Vieta's Formulas for Cubics
7	Sep 27–Oct 1	Rational & Integer Roots
8	Oct 4–Oct 8	Coefficient Tricks
9	Oct 11–Oct 15	Nonlinear Systems
10	Oct 18–Oct 22	Review & Mini-Contest 1
11	Oct 25–Oct 29	Exponential Equations
12	Nov 1–Nov 5	Logarithm Rules
13	Nov 8–Nov 12	Logarithmic Equations
14	Nov 15–Nov 19	Growth, Decay & Interest
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Big Numbers: Digits & Logs
16	Dec 6–Dec 10	Infinite Series
17	Dec 13–Dec 17	Recursive Sequences
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Complex Numbers
19	Jan 10–Jan 14	Powers of i & Roots of Unity
20	Jan 17–Jan 21	Mid-Year Mock Contest & Review
21	Jan 24–Jan 28	Trigonometry & the Unit Circle
22	Jan 31–Feb 4	Trig Identities
23	Feb 7–Feb 11	Laws of Sines & Cosines
24	Feb 14–Feb 18	Trig Area & Trig Equations
25	Feb 21–Feb 25	Circles & Parabolas
26	Feb 28–Mar 3	Ellipses & Hyperbolas
27	Mar 6–Mar 10	The Binomial Theorem
28	Mar 13–Mar 17	Advanced Counting
29	Mar 20–Mar 24	Probability & Expected Value
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Number Theory for AMC 10
31	Apr 10–Apr 14	Inequalities & Optimization
32	Apr 17–Apr 21	Contest Strategy: AMC 10 & AMC 12 Preview
33	Apr 24–Apr 28	Full Mock Contest & Review
34	May 1–May 5	Final Contest & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Functions & Graphs

Problem-Solving Toolkit for AMC 10

Core: Simplify first, look for symmetry, and test small cases.

Mon Aug 16

Look for structure

TRY Compute $\dfrac{2027^2 - 2025^2}{4}$.

Tue Aug 17

Test small cases

TRY If $1 + 3 + 5 + \cdots + (2n-1) = 400$, find n.

Wed Aug 18

Clever substitution

TRY If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.

Thu Aug 19

Step by step

TRY If $f(x) = 2x + 3$, find $f(f(f(1)))$.

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Functions & Graphs

Functions: Domain, Range & Composition

Core: Know what can go in, what comes out, and what happens when functions are chained.

Mon Aug 23

Domain

TRY Find the domain of $f(x) = \sqrt{6 - 2x}$.

Tue Aug 24

Range

TRY Find the range of $f(x) = x^2 - 4x + 7$.

Wed Aug 25

Composition

TRY If $f(x) = x^2 + 1$ and $g(x) = x - 2$, solve $f(g(x)) = 10$.

Thu Aug 26

Functional equations

TRY If $f(x) + 2f(1 - x) = 3x$ for every x, find $f(0)$.

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Functions & Graphs

Inverses & Graph Transformations

Core: Swap x and y to invert; shifts and stretches move graphs.

Mon Aug 30

Inverse values

TRY If $f(x) = \dfrac{2x+1}{x-3}$, find $f^{-1}(5)$.

Tue Aug 31

Inverse formulas

TRY Find $f^{-1}(x)$ for $f(x) = 3x - 7$.

Wed Sep 1

Shifting graphs

TRY Shift $y = x^2$ right 3 and up 2. Write the new equation.

Thu Sep 2

Vertex form

TRY Find the vertex of $y = -2(x+1)^2 + 5$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Functions & Graphs

Absolute Value & Piecewise Functions

Core: Split into cases where the expression inside changes sign.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Absolute value graphs

TRY Find the area enclosed by $y = |x|$ and $y = 4$.

Tue Sep 7

Piecewise functions

TRY $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Solve $f(x) = 4$.

Wed Sep 8

Sums of absolute values

TRY Solve $|x - 2| + |x + 3| = 7$.

Thu Sep 9

Diamond regions

TRY Find the area of the region $|x| + |y| \leq 3$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Polynomials

Remainder & Factor Theorems — **Core:** $P(a)$ is the remainder when $P(x)$ is divided by $x - a$.

Mon Sep 13
Remainder theorem
TRY Find the remainder when $x^3 - 2x^2 + 5$ is divided by $x - 3$.

Tue Sep 14
Factor theorem
TRY Find k so that $x - 2$ is a factor of $x^3 + kx - 6$.

Wed Sep 15
Polynomial division
TRY Divide $x^3 - 1$ by $x - 1$.

Thu Sep 16
Two remainders
TRY $P(x)$ leaves remainder 3 when divided by $x - 1$ and 7 when divided by $x - 3$. Find the remainder when it is divided by $(x - 1)(x - 3)$.

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Polynomials

Vieta's Formulas for Cubics — **Core:** For $x^3 + bx^2 + cx + d$: roots add to $-b$, pair products add to c, product is $-d$.

Mon Sep 20
Sum & product
TRY Find the sum and product of the roots of $x^3 - 6x^2 + 11x - 6 = 0$.

Tue Sep 21
Sum of squares
TRY r, s, t are the roots of $x^3 - 4x^2 + x + 6 = 0$. Find $r^2 + s^2 + t^2$.

Wed Sep 22
Reciprocal sums
TRY For the same roots, find $\frac{1}{r} + \frac{1}{s} + \frac{1}{t}$.

Thu Sep 23
Finding the roots
TRY Factor $x^3 - 6x^2 + 11x - 6$ completely.

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Polynomials

Rational & Integer Roots — **Core:** A rational root $\frac{p}{q}$ has p dividing the constant and q dividing the leading coefficient.

Mon Sep 27
Rational Root Theorem
TRY Find all roots of $2x^3 - 3x^2 - 3x + 2 = 0$.

Tue Sep 28
Integer roots
TRY For how many integers k does $x^2 + kx + 12 = 0$ have two integer roots?

Wed Sep 29
Building a cubic
TRY A monic cubic has roots 1, 2 and -3. Find its constant term.

Thu Sep 30
Evaluating products
TRY If $P(x) = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5)$, find $P(0)$.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Polynomials

Coefficient Tricks — **Core:** Plug in $x = 1$ or $x = -1$ to add up coefficients.

Mon Oct 4
Sum of coefficients
TRY Find the sum of the coefficients of $(2x - 1)^7$.

Tue Oct 5
Even coefficients
TRY Find the sum of the coefficients of the even powers of x in $(x + 1)^6$.

Wed Oct 6
Constant terms
TRY Find the constant term of $\left(x + \frac{2}{x}\right)^6$.

Thu Oct 7
Spot the pattern
TRY If $P(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$, find $P(11)$.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Polynomials

Nonlinear Systems **Core:** Add, subtract or multiply equations — or use $x+y$ and $x\cdot y$ as the unknowns.

Mon Oct 11
Sum & product
TRY If $x+y=7$ and $x\cdot y=12$, find x^2+y^2.

Tue Oct 12
Line meets parabola
TRY Where does $y=x+2$ meet $y=x^2$?

Wed Oct 13
Symmetric systems
TRY If $x^2+y^2=20$ and $x\cdot y=8$, find $(x+y)^2$.

Thu Oct 14
Multiply the equations
TRY Positive numbers satisfy $x\cdot y=6$, $y\cdot z=12$, $x\cdot z=8$. Find $x\cdot y\cdot z$.

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Polynomials

Review & Mini-Contest 1 **Core:** Pull Units 1–2 together and practice contest habits.

Mon Oct 18
Review: composition
TRY If $f(x)=2x-1$ and $g(x)=x^2$, find $g(f(3))-f(g(3))$.

Tue Oct 19
Review: remainders
TRY Find the remainder when x^{100} is divided by $x+1$.

Wed Oct 20
Review: Vieta
TRY Find the sum of the reciprocals of the roots of $2x^2-7x+3=0$.

Thu Oct 21
Timed set: 10 problems in 20 min
TRY If $x+y=5$ and $x^2+y^2=13$, find $x\cdot y$.

Fri Oct 22
Mini-Contest 1
Units 1–2, timed. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Exponentials & Logarithms

Exponential Equations **Core:** Rewrite with a common base, or substitute $u=a^x$.

Mon Oct 25
Common base
TRY Solve $9^{x+1}=27^x$.

Tue Oct 26
Quadratics in disguise
TRY Solve $4^x-6\cdot 2^x+8=0$.

Wed Oct 27
Comparing sizes
TRY Order from least to greatest: 2^{30}, 3^{20}, 5^{10}.

Thu Oct 28
Exponent puzzles
TRY If $2^a=3$ and $3^b=8$, find $a\cdot b$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Exponentials & Logarithms

Logarithm Rules **Core:** A logarithm is an exponent; logs turn products into sums.

Mon Nov 1
The definition
TRY Compute $\log_2 64+\log_3 \frac{1}{9}$.

Tue Nov 2
Product rule
TRY Compute $\log_6 4+\log_6 9$.

Wed Nov 3
Change of base
TRY Compute $\log_4 8$.

Thu Nov 4
Log chains
TRY Compute $\log_2 3\cdot\log_3 4\cdot\log_4 5\cdots\log_7 8$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Exponentials & Logarithms

Logarithmic Equations — **Core:** Combine logs, rewrite as an exponent, and always check the domain.

Mon Nov 8
Solve by definition
TRY Solve $\log_3(2x+1)=4$.

Tue Nov 9
Combine logs
TRY Solve $\log_2 x+\log_2(x-2)=3$.

Wed Nov 10
Check the domain
TRY Solve $\log x+\log(x-3)=1$ (base 10).

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Changing the base
TRY If $\log_2 x=5$, find $\log_4 x$.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Exponentials & Logarithms

Growth, Decay & Interest — **Core:** Repeated percent change multiplies.

Mon Nov 15
Doubling
TRY Bacteria double every 3 hours. Starting with 500, how many are there after 15 hours?

Tue Nov 16
Half-life
TRY A substance has a half-life of 8 years. What fraction is left after 24 years?

Wed Nov 17
Compound interest
TRY $1,000 earns 10% per year, compounded yearly. How much is there after 2 years?

Thu Nov 18
Percent chains
TRY A price rises 20% and then falls 20%. What is the net percent change?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Exponentials & Logarithms

Big Numbers: Digits & Logs — **Core:** A positive integer N has $\lfloor \log_{10} N \rfloor + 1$ digits.

Mon Nov 29
Counting digits
TRY Using $\log_{10} 2 \approx 0.301$, how many digits does 2^{20} have?

Tue Nov 30
Pairing 2s and 5s
TRY How many digits does $5^{30} \times 2^{25}$ have?

Wed Dec 1
Comparing with logs
TRY Which is larger: 10^{30} or 2^{100}?

Thu Dec 2
Trailing zeros
TRY How many zeros are at the end of $100!$?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Sequences, Series & Complex Numbers

Infinite Series — **Core:** A geometric series with $|r|<1$ adds up to $\frac{a}{1-r}$.

Mon Dec 6
Infinite geometric series
TRY Find $1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\cdots$.

Tue Dec 7
Repeating decimals
TRY Write $0.\overline{27}$ as a fraction in lowest terms.

Wed Dec 8
Finding the ratio
TRY An infinite geometric series has first term 6 and sum 15. Find the ratio.

Thu Dec 9
Mixed series
TRY Find $\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\cdots$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Sequences, Series & Complex Numbers

Recursive Sequences **Core:** Write out the first few terms and look for a pattern or a cycle.

Mon Dec 13
Recursion
TRY If $a_1 = 2$ and $a_{n+1} = 3a_n - 1$, find a_5.

Tue Dec 14
Cycles
TRY If $a_1 = 2$ and $a_{n+1} = \dfrac{1}{1 - a_n}$, find a_{2027}.

Wed Dec 15
Fibonacci-style counting
TRY How many ways can you climb 8 stairs taking 1 or 2 steps at a time?

Thu Dec 16
Closed forms
TRY If $a_1 = 1$ and $a_n = a_{n-1} + 2n - 1$, find a_{10}.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Sequences, Series & Complex Numbers

Complex Numbers **Core:** $i^2 = -1$: treat i like a variable, then simplify.

Mon Jan 3
Multiplying
TRY Compute $(3 + 2i)(1 - 4i)$.

Tue Jan 4
Dividing
TRY Compute $\dfrac{5 + 5i}{1 + 2i}$.

Wed Jan 5
Modulus
TRY Find $|6 - 8i|$.

Thu Jan 6
Complex roots
TRY Solve $x^2 + 4x + 13 = 0$.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Sequences, Series & Complex Numbers

Powers of i & Roots of Unity **Core:** Powers of i repeat every 4; roots of unity are evenly spaced on a circle.

Mon Jan 10
Powers of i
TRY Find i^{2027}.

Tue Jan 11
Sums of powers
TRY Find $i + i^2 + i^3 + \cdots + i^{100}$.

Wed Jan 12
Repeated squaring
TRY Find $(1 + i)^8$.

Thu Jan 13
Cube roots of unity
TRY If $\omega^3 = 1$ and $\omega \neq 1$, find $1 + \omega + \omega^2$.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Sequences, Series & Complex Numbers

Mid-Year Mock Contest & Review **Core:** Review Units 1–4 and practice pacing a full contest.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: logarithms
TRY Compute $\log_2 48 - \log_2 3$.

Tue Jan 18
Review: series
TRY An infinite geometric series has sum 8 and ratio $\frac{1}{4}$. Find the first term.

Wed Jan 19
Review: complex numbers
TRY Compute $(2 - i)^2$.

Thu Jan 20
Pacing: skip & return
TRY 15-minute set. Warm-up: find the remainder when $x^3 + 1$ is divided by $x - 2$.

Fri Jan 21
Mid-Year Mock Contest
Full-length, Units 1–4. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Trigonometry & Conics

Trigonometry & the Unit Circle — **Core:** SOH-CAH-TOA in right triangles, then extend to any angle with the unit circle.

Mon Jan 24
SOH-CAH-TOA
TRY A right triangle has legs 5 and 12. Find the sine of the angle opposite the side 5.

Tue Jan 25
Special angles
TRY Compute $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ$.

Wed Jan 26
The unit circle
TRY Find $\cos 150^\circ$.

Thu Jan 27
Pythagorean identity
TRY If $\sin\theta = \frac{3}{5}$ and θ is obtuse, find $\cos\theta$.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Trigonometry & Conics

Trig Identities — **Core:** $\sin^2\theta + \cos^2\theta = 1$, plus the sum and double-angle formulas.

Mon Jan 31
Double angle: sine
TRY If $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$, find $\sin 2\theta$.

Tue Feb 1
Double angle: cosine
TRY If $\cos\theta = \frac{1}{3}$, find $\cos 2\theta$.

Wed Feb 2
Sum formulas
TRY Find the exact value of $\sin 75^\circ$.

Thu Feb 3
Simplifying
TRY Simplify $(\sin x + \cos x)^2 - \sin 2x$.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Trigonometry & Conics

Laws of Sines & Cosines — **Core:** Use cosines for SAS or SSS; use sines for side–angle pairs.

Mon Feb 7
Law of cosines
TRY A triangle has sides 5 and 8 with a 60° angle between them. Find the third side.

Tue Feb 8
Finding an angle
TRY A triangle has sides 7, 8 and 13. Find its largest angle.

Wed Feb 9
Law of sines
TRY In $\triangle ABC$, $\angle A = 30^\circ$ and $a = 6$. Find the circumradius.

Thu Feb 10
The ambiguous case
TRY How many triangles have $\angle A = 30^\circ$, $a = 5$ and $b = 8$?

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Trigonometry & Conics

Trig Area & Trig Equations — **Core:** Area $= \frac{1}{2}ab\sin C$; solve trig equations with the unit circle.

Mon Feb 14
Area with sine
TRY A triangle has sides 6 and 10 with a 30° angle between them. Find its area.

Tue Feb 15
Basic equations
TRY Solve $2\sin x = 1$ for $0^\circ \le x < 360^\circ$.

Wed Feb 16
Quadratic trig equations
TRY Solve $2\cos^2 x - \cos x - 1 = 0$ for $0^\circ \le x < 360^\circ$.

Thu Feb 17
Regular polygons
TRY Find the area of a regular octagon inscribed in a circle of radius 2.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Trigonometry & Conics

Circles & Parabolas — **Core:** Complete the square to read off centers, radii and vertices.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Completing the square

TRY Find the center and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

Tue Feb 22

Tangent lines

TRY Is the line $y = x + 4$ tangent to the circle $x^2 + y^2 = 8$?

Wed Feb 23

Vertex

TRY Find the vertex of $y = 2x^2 - 8x + 3$.

Thu Feb 24

Focus

TRY Find the focus of the parabola $y = \frac{x^2}{8}$.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Trigonometry & Conics

Ellipses & Hyperbolas — **Core:** Ellipse: the distances to the foci add to a constant; hyperbola: they differ by one.

Mon Feb 28

Ellipse axes

TRY Find the lengths of the major and minor axes of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

Tue Feb 29

Foci

TRY Find the foci of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

Wed Mar 1

Ellipse area

TRY An ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has area πab. Find the area of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

Thu Mar 2

Asymptotes

TRY Find the asymptotes of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Counting, Probability & Number Theory

The Binomial Theorem — **Core:** The coefficients of $(a + b)^n$ come from Pascal's triangle.

Mon Mar 6

Coefficients

TRY Find the coefficient of x^3 in $(x + 3)^5$.

Tue Mar 7

Middle term

TRY Find the middle term of $(x - 2)^6$.

Wed Mar 8

Row sums

TRY Find $\binom{10}{0} + \binom{10}{1} + \cdots + \binom{10}{10}$.

Thu Mar 9

Hockey stick identity

TRY Find $\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \cdots + \binom{8}{3}$.

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Counting, Probability & Number Theory

Advanced Counting — **Core:** Inclusion–exclusion, complements and clever one-to-one matchings.

Mon Mar 13

Inclusion–exclusion

TRY How many integers from 1 to 1,000 are divisible by 2, 3 or 5?

Tue Mar 14

Derangements

TRY Four letters go into four addressed envelopes, none in the right one. How many ways?

Wed Mar 15

Stars & bars

TRY How many solutions in positive integers does $a + b + c = 10$ have?

Thu Mar 16

Circular arrangements

TRY How many ways can 6 people sit around a round table (rotations count as the same)?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Counting, Probability & Number Theory

Probability & Expected Value

Core: Expected values add; probabilities of independent events multiply.

Mon Mar 20
Conditional probability
TRY A family has two children, and at least one is a girl. What is the probability both are girls?

Tue Mar 21
Expected value
TRY What is the expected number of heads in 10 fair coin flips?

Wed Mar 22
Linearity
TRY Two dice are rolled. What is the expected sum?

Thu Mar 23
Waiting time
TRY On average, how many rolls of a die does it take to get the first 6?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Counting, Probability & Number Theory

Number Theory for AMC 10

Core: Work with remainders, divisors and integer equations.

Mon Apr 3
Modular inverses
TRY Find x with $0 \le x < 7$ so that $3x$ leaves remainder 1 when divided by 7.

Tue Apr 4
Integer equations
TRY How many solutions in positive integers does $5x + 7y = 100$ have?

Wed Apr 5
Sum of divisors
TRY Find the sum of all positive divisors of 36.

Thu Apr 6
Powers in factorials
TRY What is the largest n for which 2^n divides $20!$?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Counting, Probability & Number Theory

Inequalities & Optimization

Core: AM–GM: the average of positive numbers is at least their geometric mean.

Mon Apr 10
AM–GM
TRY Find the smallest value of $x + \frac{9}{x}$ for $x > 0$.

Tue Apr 11
Fixed sum
TRY Positive x and y add to 10. Find the largest possible value of $x \cdot y$.

Wed Apr 12
Completing the square
TRY Find the smallest value of $x^2 - 6x + 14$.

Thu Apr 13
Fixed product
TRY Positive a and b satisfy $a \cdot b = 16$. Find the smallest value of $a + 4b$.

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Contest Season Wrap-Up

Contest Strategy: AMC 10 & AMC 12 Preview

Core: Budget your time, skip wisely, and use the answer choices.

Mon Apr 17
Quick computation
TRY Compute $\sqrt{16+9} + \sqrt[3]{27}$.

Tue Apr 18
Clever substitution
TRY If $a + b = 3$ and $a^3 + b^3 = 9$, find $a \cdot b$.

Wed Apr 19
AMC 12 style
TRY Solve $\log_2(\log_3(\log_4 x)) = 0$.

Thu Apr 20
Timed set: 10 in 20 min
TRY The number 2027 is prime. How many positive divisors does 2027^2 have?

Fri Apr 21
Weekly Quiz
10 mixed contest problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Contest Season Wrap-Up

Full Mock Contest & Review **Core:** Simulate contest day and learn from every error.

Mon Apr 24
Review: polynomials
TRY Find the sum of the roots of $x^4 - 5x^3 + 2x - 7 = 0$.

Tue Apr 25
Review: trigonometry
TRY Compute $\cos^2 15^\circ - \sin^2 15^\circ$.

Wed Apr 26
Review: logarithms
TRY Compute $\log_3 81 - \log_2 \frac{1}{8}$.

Thu Apr 27
Full Mock Contest
Full-length mixed contest under real time limits.

Fri Apr 28
Error-log review
Redo every missed problem from the mock; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Contest Season Wrap-Up

Final Contest & Celebration **Core:** Show what you have learned and set next year's goals.

Mon May 1
Light review: favorite strategies
TRY Compute $i^4 + i^8 + i^{12}$.

Tue May 2
Mixed warm-up set
TRY Find $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$.

Wed May 3
Final Contest
Full-length contest covering the whole year.

Thu May 4
Contest review
Go over solutions together; students present one problem each.

Fri May 5
Celebration & reflection
Certificates, progress review, and goals for next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 2026
Tue: 20
Wed: 18
Thu: 29

Week 2
Mon: $x \le 3$
Tue: $f(x) \ge 3$
Wed: $x = 5$ or $x = -1$
Thu: 2

Week 3
Mon: $\frac{16}{3}$
Tue: $f^{-1}(x) = \frac{x+7}{3}$
Wed: $y = (x-3)^2 + 2$
Thu: $(-1, 5)$

Week 4
Mon: 16
Tue: $x = 2$ only
Wed: $x = 3$ or $x = -4$
Thu: 18

Week 5
Mon: 14
Tue: $k = -1$
Wed: $x^2 + x + 1$
Thu: $2x + 1$

Week 6
Mon: Sum 6, product 6
Tue: 14
Wed: $-\frac{1}{6}$
Thu: $(x-1)(x-2)(x-3)$

Week 7
Mon: $2, -1, \frac{1}{2}$
Tue: 6
Wed: 6
Thu: -120

Week 8
Mon: 1
Tue: 32
Wed: 160
Thu: 10,000

Week 9
Mon: 25
Tue: $(2, 4)$ and $(-1, 1)$
Wed: 36
Thu: 24

Week 10
Mon: 8
Tue: 1
Wed: $\frac{7}{3}$
Thu: 6

Week 11
Mon: $x = 2$
Tue: $x = 1$ or $x = 2$
Wed: $5^{10} < 2^{30} < 3^{20}$
Thu: 3

Week 12
Mon: 4
Tue: 2
Wed: $\frac{3}{2}$
Thu: 3

Week 13
Mon: $x = 40$
Tue: $x = 4$
Wed: $x = 5$ ($x = -2$ is rejected)
Thu: $\frac{5}{2}$

Week 14
Mon: 16,000
Tue: $\frac{1}{8}$
Wed: \$1,210
Thu: A 4% decrease

Week 15
Mon: 7
Tue: 29
Wed: 2^{100}
Thu: 24

Week 16
Mon: $\frac{3}{2}$
Tue: $\frac{3}{11}$
Wed: $\frac{3}{5}$
Thu: 2

Week 17
Mon: 122
Tue: -1
Wed: 34
Thu: 100

Week 18
Mon: $11 - 10i$
Tue: $3 - i$
Wed: 10
Thu: $x = -2 \pm 3i$

Week 19
Mon: $-i$
Tue: 0
Wed: 16
Thu: 0

Week 20
Mon: 4
Tue: 6
Wed: $3 - 4i$
Thu: 9

Week 21
Mon: $\frac{5}{13}$
Tue: 2
Wed: $-\frac{\sqrt{3}}{2}$
Thu: $-\frac{4}{5}$

Week 22
Mon: $\frac{24}{25}$
Tue: $-\frac{7}{9}$
Wed: $\frac{\sqrt{6}+\sqrt{2}}{4}$
Thu: 1

Week 23
Mon: 7
Tue: $120°$
Wed: 6
Thu: 2

Week 24
Mon: 15
Tue: $30°, 150°$
Wed: $0°, 120°, 240°$
Thu: $8\sqrt{2}$

Week 25
Mon: Center $(3, -2)$, radius 5
Tue: Yes: its distance from the center is $2\sqrt{2}$
Wed: $(2, -5)$
Thu: $(0, 2)$

Week 26
Mon: 10 and 6
Tue: $(\pm 4, 0)$
Wed: 15π
Thu: $y = \pm\frac{3}{4}x$

Week 27
Mon: 90
Tue: $-160x^3$
Wed: 1,024
Thu: 126

Week 28
Mon: 734
Tue: 9
Wed: 36
Thu: 120

Week 29
Mon: $\frac{1}{3}$
Tue: 5
Wed: 7
Thu: 6

Week 30
Mon: 5
Tue: 2
Wed: 91
Thu: 18

Week 31
Mon: 6
Tue: 25
Wed: 5
Thu: 16

Week 32
Mon: 8
Tue: 2
Wed: $x = 64$
Thu: 3

Week 33
Mon: 5
Tue: $\frac{\sqrt{3}}{2}$
Wed: 7

Week 34
Mon: 3
Tue: 2

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